

Performance Testing

Date	15 March 2026
Team ID	xxxxxx
Project Name	Smart Lender-Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Loan Testing
Target Module / API	Loan Prediction Module(/predict)
Test Environment	Local System(Flask Application)
Test Date	14 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user loan prediction	1	30 sec	Prediction displayed
2	Concurrent request	20	60 sec	All requests processed
3	Heavy load test	50	120 sec	Stable response

4	Continuous requests	100	180 sec	No server crash
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Meets target
2	Response Time (Max)	< 5 seconds	3.4 sec	Pass	Stable
3	Throughput (Req/sec)	20 req/sec	24 req/sec	Pass	Good
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	52%	Pass	Normal
6	Memory Utilization	< 80%	61%	pass	Normal

Step 4: Observations & Analysis

Key Findings: The Smart Lender application handled expected user load successfully. Loan predictions were generated quickly with no failure.

Bottlenecks Identified: minor increase in response time under higher concurrent load, but within acceptable limits.

Optimization Steps taken: Optimized Random Forest model loading, efficient preprocessing, and flask request handling

Step 5: Screenshots / Evidence

Attach screenshots of Apache JMeter or Application output here before final submission

Smart Lender

AI Based Loan Approval Prediction System

Gender

Male

Married

Yes

Dependents

0

Education

Graduate

Self Employed

Yes

Applicant Income

Co Applicant Income

Loan Amount

Loan Term

Credit History

Good

Property Area

Urban

Predict Loan

Smart Lender Prediction

Loan Approved 

Confidence Score

52.0%

Check Another Application